

CHAPTER 02

Energy

It powers your phone, your body, your music, and every star in the universe. It's never made, never destroyed — only transformed. Welcome to the most important currency in all of physics.

⚡ Georgia Standard S8P2

WHAT YOU'LL LEARN

- | | |
|-------------------------------|--------------------------------------|
| 01 What Is Energy? | 06 Energy Transformations Everywhere |
| 02 The Many Forms of Energy | 07 The Law of Conservation of Energy |
| 03 Kinetic Energy: The Math | 08 Heat Energy & Temperature |
| 04 Potential Energy: The Math | 09 Three Types of Heat Transfer |
| 05 KE ↔ PE Transformations | |

2.1 What Is Energy?

You charge your phone. You eat a sandwich. You scroll through an app powered by satellites circling Earth. All of these have one thing in common: **energy**. Energy is the **ability to do work or cause change**. Without it, nothing happens — not a single thing in the universe.

What does "doing work" mean? In science, **work** happens when a force makes something move. Pushing a shopping cart forward is work. Lifting your backpack onto a desk is work. Sitting still and staring at the cart? Not work — no matter how tired you get holding it.

The unit scientists use to measure energy is the **joule** (abbreviated J), named after the 19th-century scientist James Prescott Joule. One joule is roughly the energy needed to lift a small apple about 1 meter into the air.

! Did You Know?

The food on your plate is rated in **Calories** — and 1 food Calorie equals about **4,184 joules**. That bag of hot chips with 200 Calories? It contains about **837,000 joules** of energy. Your body either uses that energy to move and grow, or stores it for later.

X Common Mix-Up

Energy isn't "stuff." You can't put energy in a bucket. It's not matter — it has no mass and takes up no space. Energy is more like a *property* that objects have, similar to how something can be "fast" or "hot."

2.2 The Many Forms of Energy

Energy comes in lots of different forms. The big categories you need to know are:

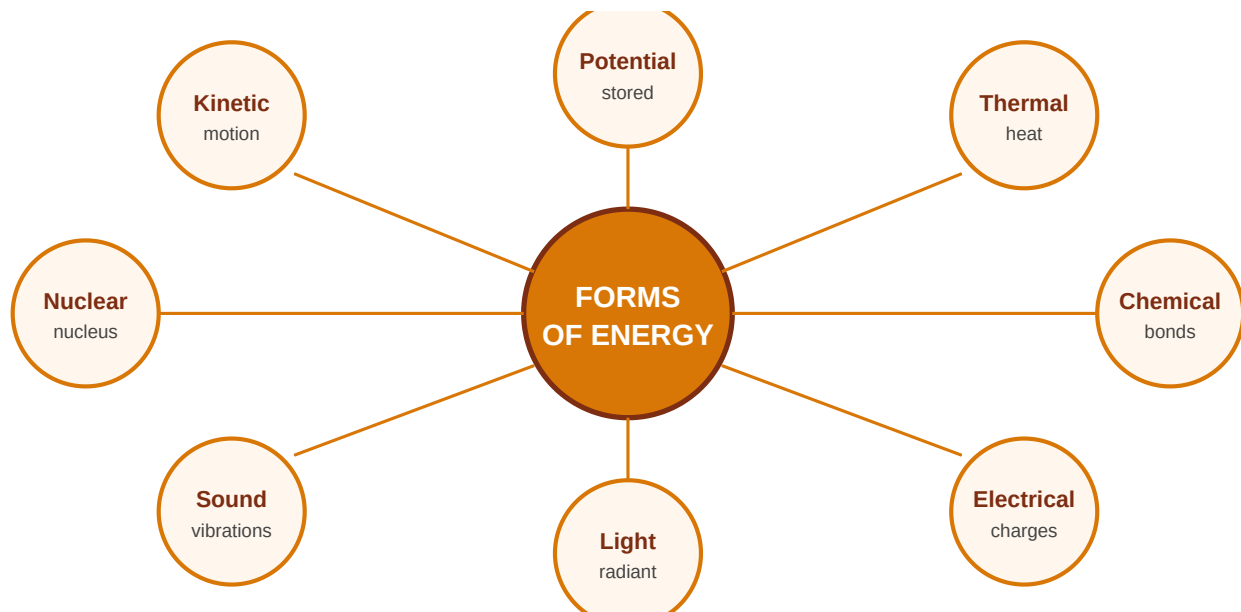


Figure 2.1 — The major forms of energy. They all do the same thing (cause change), just in different ways.

Form	What It Is	Real Example
Kinetic	The energy of motion	A bike rolling downhill, a soccer ball in the air
Potential	Stored energy due to position or condition	A book on a high shelf, a stretched rubber band
Mechanical	The total of an object's KE + PE	A swinging pendulum, a moving car
Thermal (Heat)	Energy from particle motion in matter	A hot cup of coffee, a campfire
Chemical	Energy stored in the bonds between atoms	Food, gasoline, batteries, fireworks
Electrical	Energy from moving electric charges	Lightning, current in your phone charger
Light (Radiant)	Energy carried by electromagnetic waves	Sunlight, glow of a screen
Sound	Energy carried by vibrating particles	Your voice, music from a speaker
Nuclear	Energy stored inside an atom's nucleus	The Sun's core, nuclear power plants
Elastic	Stored energy in a stretched/compressed object	A drawn bow, a coiled spring, a trampoline

! Did You Know?

That sandwich at lunch had **chemical energy** stored in its molecular bonds. Your body's digestive system breaks those bonds, releasing that energy so you can **walk, think, and breathe**. Every step you take is powered by chemistry that started millions of years ago when plants absorbed light energy from the Sun. Your lunch is literally bottled sunshine.

? Quick Check 2.2

1. Identify the form of energy in each: (a) a phone battery, (b) a kid sliding down a slide, (c) the Sun, (d) a TV remote signal, (e) a slingshot pulled back.
2. Why is a book on a tall shelf said to have *more* energy than a book on a low shelf?

2.3 Kinetic Energy: The Energy of Motion

Kinetic energy (KE) is the energy an object has because it's **moving**. Anything in motion — a baseball flying through the air, a car driving down I-575, an electron zipping around an atom — has kinetic energy.

Two factors affect how much kinetic energy something has:

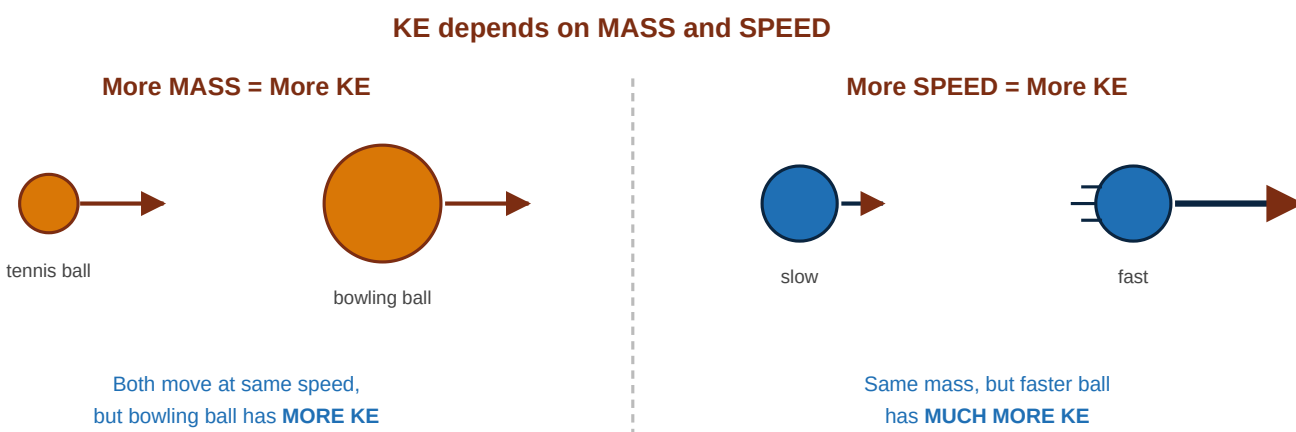


Figure 2.2 — Both mass and speed increase kinetic energy. But speed has a bigger effect because of the math!

★ The KE Formula

$$KE = \frac{1}{2} \times m \times v^2$$

The ² on the speed (v) means speed has a **HUGE** effect on KE. Double the speed = **4×** the KE. Triple the speed = **9×** the KE. This is why crashes at high speeds are so much more dangerous than at low speeds.

If you graphed mass vs. KE (with speed held constant), you'd get a **straight line** — KE grows steadily as mass increases. But if you graph speed vs. KE (with mass held constant), you get a **curve** that shoots upward fast, because of that squared term.

? Quick Check 2.3

1. Two cars have the same mass. Car A drives 30 mph; Car B drives 60 mph. Which has more KE? About how many times more?
2. An elephant walks slowly; a mouse sprints. Which one might have more KE? What does it depend on?

2.4 Potential Energy: Stored & Ready

Potential energy (PE) is energy that is **stored** in an object because of its position or condition. It hasn't been released yet — but it's ready to be.

Think of a roller coaster sitting at the top of the first hill before the drop. It isn't moving. But every passenger knows that energy is about to be UNLEASHED. That's potential energy.

Gravitational Potential Energy

The most common type in 8th grade physics is **gravitational potential energy** — stored energy due to an object's height above the ground. It depends on two factors:

Gravitational PE depends on MASS and HEIGHT

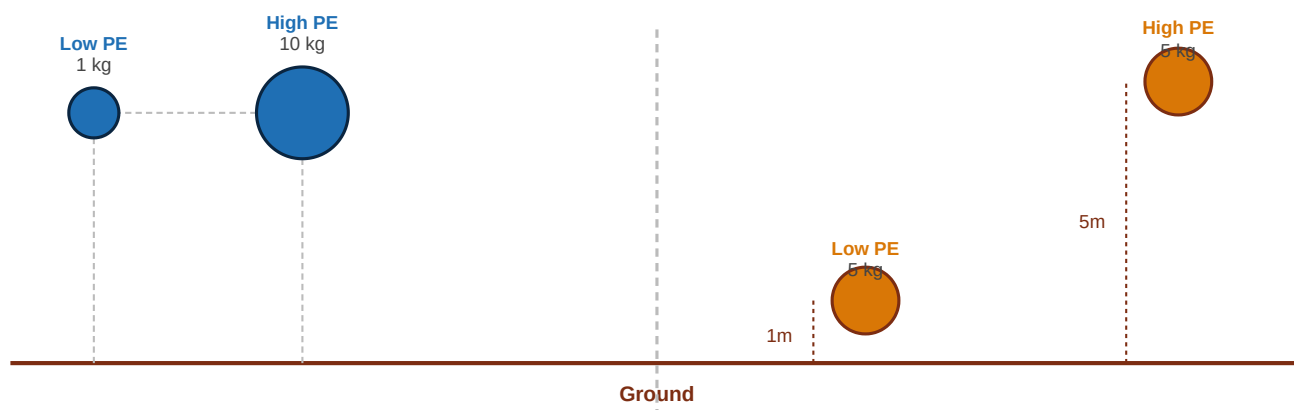


Figure 2.3 — PE goes up when you increase either mass or height. Unlike KE, both relationships are straight lines.

★ The PE Formula

$$PE = m \times g \times h$$

Where m = mass, g = gravity (about 9.8 m/s² on Earth), and h = height. The simple version: heavier = more PE; higher = more PE. Both relationships are **linear** — double either one, you double the PE.

! Did You Know?

The Hoover Dam holds back so much water that the gravitational potential energy of the water behind it can be converted into about **4 billion kilowatt-hours of electricity per year** — enough to power 1.3 million homes. Water sitting still at the top has the energy. Letting it fall releases it.

2.5 KE ↔ PE Transformations

Here's where it gets cool: **kinetic and potential energy constantly trade places**. As one goes up, the other goes down. The total (the mechanical energy) stays the same.

Roller Coaster: PE and KE Trade Places

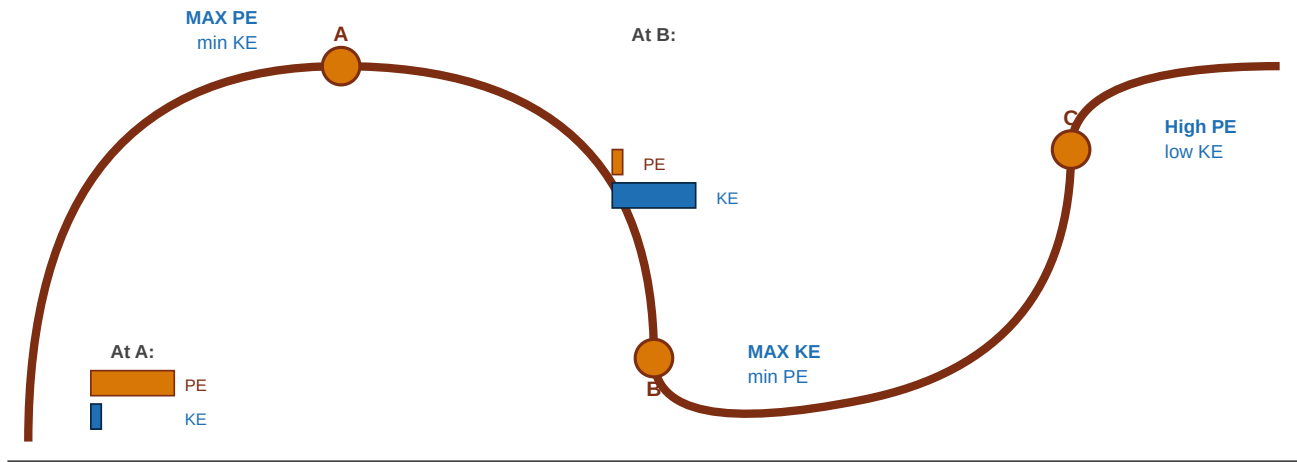


Figure 2.4 — At the top of the hill (A), almost all the energy is PE. At the bottom (B), it's all KE. The hill at C is lower, so less PE — but the coaster has slowed down, too.

Other Examples of KE ↔ PE Trades

- **Pendulum:** At the top of its swing, it pauses — all PE, no KE. At the bottom, it's moving fastest — all KE, no PE. Up the other side, back to PE.
- **Rubber Band / Slingshot:** Stretched = high elastic PE. Released = KE as it flies forward.
- **Bouncing Ball:** Falling = PE → KE. Hitting the ground = KE → elastic PE in the ball's squish. Bouncing back up = elastic PE → KE → PE.
- **Diver Jumping Off Board:** Standing on platform = max PE. Mid-fall = KE growing. Hitting water = transfers to the water.

X Common Mix-Up

People sometimes think energy "disappears" at the top of a pendulum swing because the pendulum stops moving. **It didn't disappear — it transformed.** All the KE became PE. The pendulum has just as much energy at the top of the swing as it had at the bottom; it's just stored differently.

2.6 Energy Transformations Everywhere

An **energy transformation** is a change from one form of energy to another. They happen literally everywhere, all the time. Here are some you experience every day:

Object / Action	Energy Transformation
Striking a match	Chemical → Thermal + Light
Turning on a light bulb	Electrical → Light + Thermal
Playing music on a speaker	Electrical → Sound
Charging your phone	Electrical → Chemical (stored in battery)
Eating food	Chemical → Kinetic + Thermal (for your body)
A solar panel	Light → Electrical
A microwave heating food	Electrical → Light (microwaves) → Thermal
A flashlight	Chemical (battery) → Electrical → Light
A car engine	Chemical (gas) → Thermal → Kinetic
Fireworks	Chemical → Light + Sound + Thermal

★ Real-World Connection

Trace the energy used to send a text message: **Chemical** energy in food → **Kinetic** energy of your fingers tapping → **Electrical** energy in the phone → **Radio waves** (light energy) sent to a tower → **Electrical** energy in your friend's phone → **Light** from their screen → **Light** reaching their eyes. Seven energy transformations, in less than a second!

? Quick Check 2.6

1. Identify the energy transformation: (a) a hair dryer, (b) a candle, (c) a kid pedaling a bike, (d) a wind turbine.
2. A roller coaster engine pulls the train to the top of the first hill. What energy transformations happen during the whole ride?

2.7 The Law of Conservation of Energy

This is one of the most fundamental rules in all of physics — right up there with conservation of mass:

! Law of Conservation of Energy

Energy cannot be created or destroyed. It can only be transformed from one form into another.

The total amount of energy in a closed system stays the same forever.

This means that when a roller coaster races down the hill, the PE doesn't *vanish* — it converts into KE. When a battery dies, the chemical energy didn't disappear — it became light, sound, and heat as your device used it.

So Why Do Things "Lose" Energy?

If energy is conserved, why does a ball bouncing on the floor eventually stop bouncing? Why does a swinging pendulum slow down?

The answer: **friction and air resistance**. These forces convert some of the kinetic energy into **thermal energy** (heat) and **sound**. The total energy is still the same — but some of it has been transferred out of the system in ways you can't easily see. The ball isn't losing energy; it's just transferring it to the floor, the air, and the surroundings as heat.

! Did You Know?

Ever rubbed your hands together on a cold day to warm them up? You just witnessed conservation of energy. The kinetic energy of your moving hands turned into **thermal energy** due to friction between your palms. Nothing was lost — just transformed.

2.8 Heat Energy & Temperature

Heat is probably the most common form of energy transfer you'll see in this chapter, so let's break it down carefully. Two terms get mixed up all the time:

★ Temperature

The **average kinetic energy** of the particles in a substance. How fast — on average — are the particles vibrating or moving? Measured in degrees (°C, °F, K).

★ Heat (Thermal Energy)

The **total energy** of all the moving particles in a substance. Depends on temperature AND the amount of matter. Measured in joules.

✗ Common Mix-Up

A bathtub full of warm water has **MORE thermal energy** than a cup of boiling water. The cup has higher *temperature*, but the bathtub has way more *particles*, so the total energy adds up to more. Temperature ≠ heat!

How Heat Moves

Heat always flows from **warmer to cooler** objects — never the reverse. When you grab an ice cube, your hand doesn't "get cold" — your heat is flowing INTO the ice cube. The ice cube melts because *your* thermal energy went into it.

When heat is added to a substance, its particles move faster and the temperature usually rises. When heat is removed, particles slow down and the temperature drops. (Remember the heating curves from Chapter 1? Temperature only rises during the sloped parts — during a phase change, the energy is breaking bonds instead.)

? Quick Check 2.8

1. An ice cube at 0°C and an iceberg at 0°C have the same temperature. Which has more thermal energy? Why?
2. When you stir a hot drink with a spoon, what happens to the temperature of the spoon? Which direction does heat flow?

2.9 Three Types of Heat Transfer

Heat moves in **three different ways**: by direct contact (conduction), by moving fluid (convection), or by waves through space (radiation). Let's look at each one.

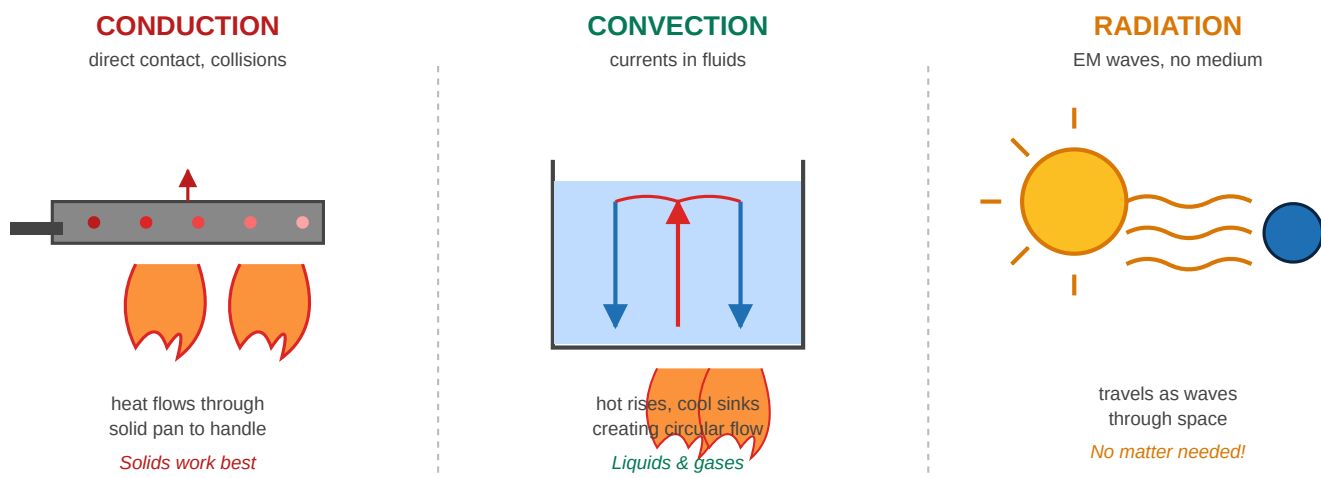


Figure 2.5 — The three modes of heat transfer. Each one moves energy differently.

1. Conduction — Heat Through Direct Contact

Conduction is the transfer of heat **through direct contact between particles**. Fast-moving particles in a hot object bump into slower particles in a cooler object, passing on their energy. This works best in **solids**, where particles are packed close together.

Examples: A metal spoon getting hot in a soup pot; your hand warming up when you touch a hot mug; a heating pad warming your back.

2. Convection — Heat Through Moving Fluids

Convection is the transfer of heat through **currents in a liquid or gas**. When a fluid is heated, its particles spread out and become less dense — so the warm fluid rises. Cooler fluid sinks to take its place, then heats up, and the cycle repeats. This creates a **convection current**.

Examples: Boiling water; hot air rising from a radiator; wind in the atmosphere; ocean currents; the bubbling motion of soup.

3. Radiation — Heat Through Empty Space

Radiation is the transfer of heat through **electromagnetic waves**. Unlike conduction and convection, radiation does **not need any matter** to travel through. That's how the Sun's heat reaches Earth across 93 million miles of empty space.

Examples: Feeling the warmth of the Sun; warmth from a campfire from a distance; the heat you feel from a glowing toaster element; a microwave heating leftovers.

★ All Three in One Place

A campfire is a great example of all three at once. Touch a metal stick that's been in the fire (**conduction**). The smoke and hot air rising are **convection**. The warmth you feel on your face when sitting near the fire is **radiation**.

? Quick Check 2.9

1. Identify the type of heat transfer: (a) a hot air balloon rising, (b) sunburn at the beach, (c) a metal pan handle getting hot, (d) warm water rising in a kettle.
2. Why can radiation travel through space but conduction cannot?
3. Why are pot handles often made of wood or plastic instead of metal?

★ CHAPTER 2 BIG IDEAS

- **Energy** is the ability to do work or cause change. Measured in **joules**.
- **Kinetic energy** (motion) depends on mass and speed ($KE = \frac{1}{2}mv^2$). Speed has a bigger effect because it's squared.
- **Potential energy** (stored) depends on mass and height ($PE = mgh$). Both have linear relationships to PE.
- KE and PE constantly trade in **roller coasters, pendulums, and rubber bands**. As one rises, the other falls.
- Energy transformations happen everywhere (chemical → light + heat, electrical → light, etc.).
- **Conservation of Energy**: energy is never created or destroyed — only transformed.
- **Temperature** is average particle KE; **thermal energy** is total particle energy. They're different!
- Heat moves three ways: **conduction** (contact, solids), **convection** (currents, fluids), and **radiation** (waves, no medium needed).

I Can... (Self-Check)

- ☐ I can analyze and create graphical displays showing the relationships of kinetic energy to mass and speed, and potential energy to mass and height. (S8P2.a)
- ☐ I can explain the transformation between kinetic and potential energy in systems like roller coasters, pendulums, and rubber bands. (S8P2.b)
- ☐ I can identify the type of energy transformation in a system (e.g., chemical → light + heat in a match, electrical → light in a bulb). (S8P2.c)
- ☐ I can explain how heat transfer affects molecular motion through conduction (collision of atoms), radiation (through space), and convection (currents in a fluid). (S8P2.d)